Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The dataset was split into 70 percent training data and the rest for testing. Each experiment was repeated 5 times with different random seeds.

2 Results

Figure 1 shows the accuracy by dataset.

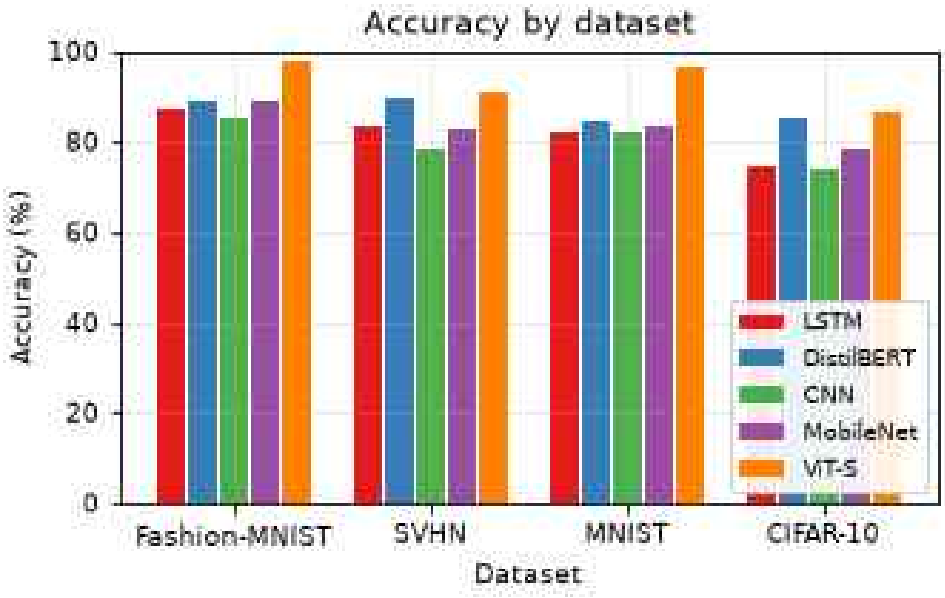


Figure 1: Accuracy by dataset for the models in this experiment.

ViT-S reaches about 90% accuracy on CIFAR-10. As shown in Figure 1, ViT-S reaches 84.7% accuracy on CIFAR-10. As shown in Figure 1, on SVHN, ViT-S obtains a accuracy of 85.6%.

Figure 2 shows the f1 score of MLP by dataset.

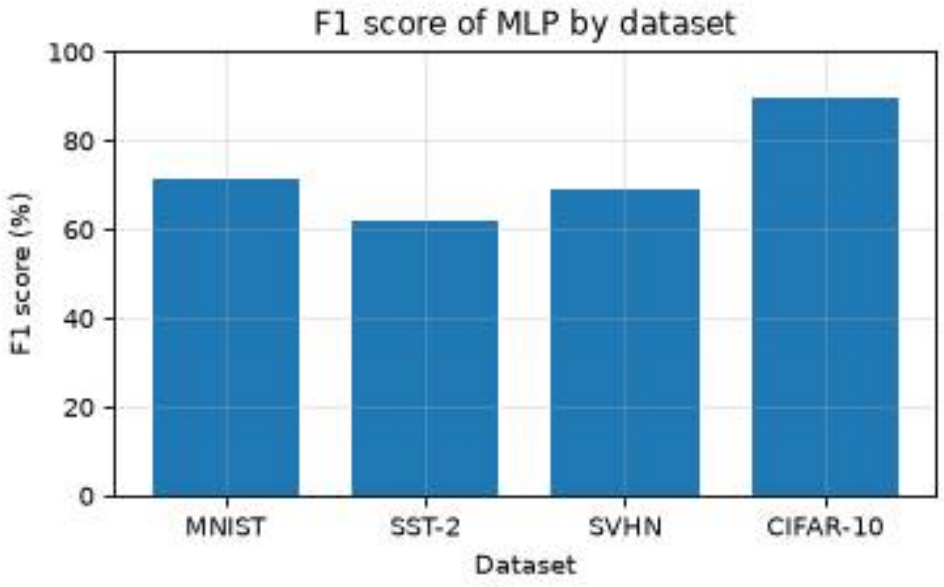


Figure 2: F1 score of MLP by dataset for the models in this experiment.

As shown in Figure 2, MLP achieves 62.2% F1 score on SST-2. On MNIST, MLP obtains a F1 score of 21.5%.

Figure 3 shows the validation accuracy during training.

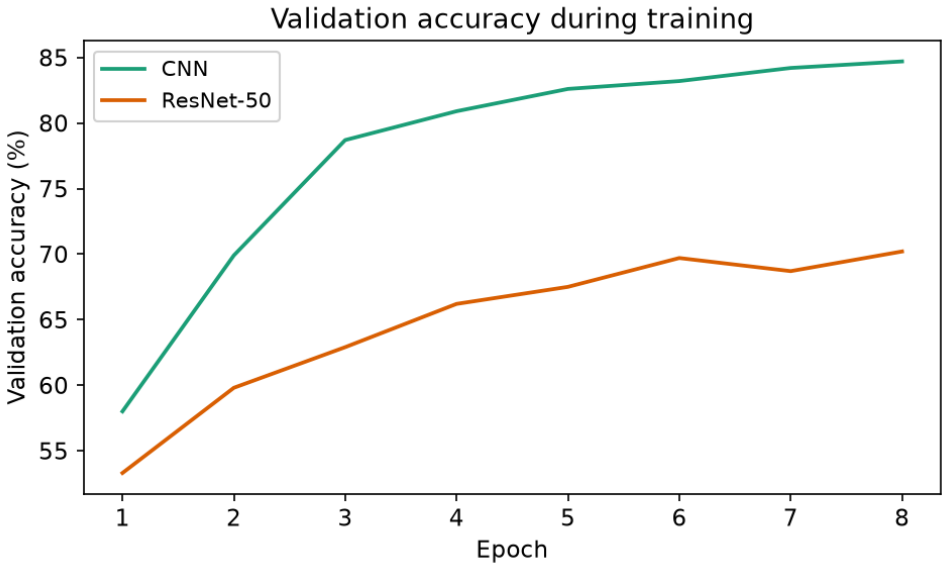


Figure 3: Validation accuracy during training for the models in this experiment.

From epoch 3 to epoch 4, the validation accuracy of ResNet-50 improves by a relative 10.7%. ResNet-50 reaches 62.8% validation accuracy at epoch 6 (Figure 3). As shown in Figure 3, at epoch 3, CNN is 16 points better than ResNet-50.

Figure 4 shows the validation accuracy during training.

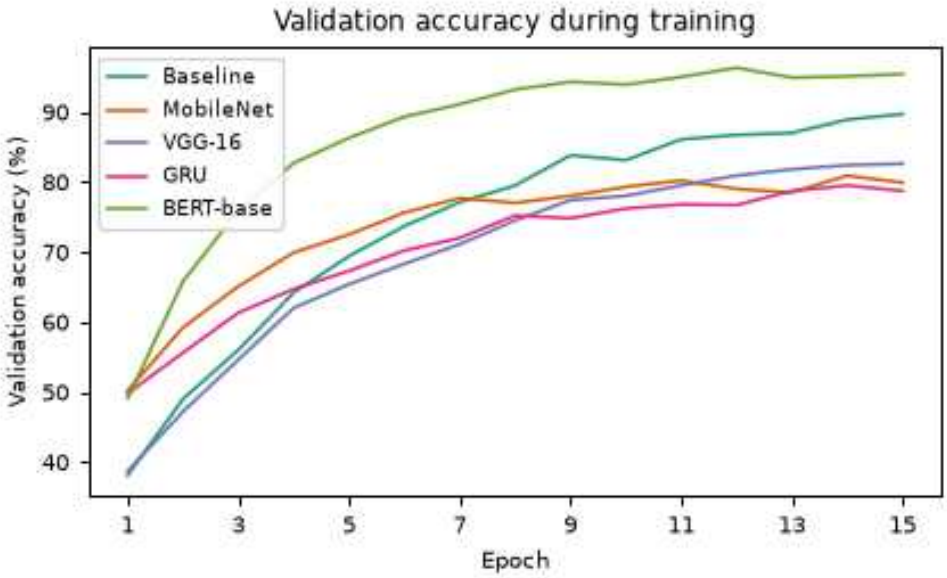


Figure 4: Validation accuracy during training for the models in this experiment.

Baseline reaches 60.9% validation accuracy at epoch 6 (Figure 4). BERT-base outperforms VGG-16 by about 16 percentage points at epoch 10.

3 Discussion

Logistic Regression reaches 86.2% accuracy on SVHN. The differences between the models are discussed in the next section.